

Gao Haiyang

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Educational Background

Imperial College London

09/2026-09/2027

Degree: Master of Science (in progress)

Major: Artificial Intelligence

Wuhan University Honor Graduate of Hongyi Honor College

09/2022-06/2026

Degree: Bachelor of Economics & Bachelor of Mathematics

Honours: Honor Graduate of Hongyi Honor College

Major: Finance

Minor: Mathematics and Applied Mathematics

Mark: 3.69/4.0 (Major)

Core Modules: (Numbers in the bracket is course grade out of 100)

Real Analysis (95), Numerical Analysis (95), Stochastic Process (91), Methods of Data Analysis (91), Machine Learning (90), Monetary Banking (96), Corporate Finance (95), Options, Futures and other Derivatives (93), Theory of Markets and Equilibrium (90)

Yale Summer Session

06/2025-08/2025

Courses: Game Theory (course grade: A), YData: Introduction to Data Science (course grade: A)

Published Papers

2024 Green Finance Annual Report, 2025, Second author, Industry Publication

Academic Projects

Sparseformer: Improving Long-Range Dependency Modeling for Time-Series Forecasting 07/2025-03/2026

Position: Researcher and first author

- Designed Sparseformer, an encoder-decoder model that combines Transformable Sparse Attention with offset-network sampling and ProbSparse global attention, expanding the effective receptive field while retaining $O(L \log L)$ computational complexity and $O((2 - \epsilon)L \log L)$ memory complexity; enriched input representations with trend, seasonal, and residual components from STL decomposition.
- Conducted PyTorch experiments on ETTh1, WTH, and ECL, benchmarking against Informer, Reformer, and DeepAR using MSE/MAE; achieved the best result on every dataset and metric. [Paper](#)

X-KANformer: A KAN-Enhanced Transformer for Portfolio Allocation 03/2026-08/2026

Position: Researcher and first author

- Led the design of the encoder-only X-KANformer, replacing the conventional Transformer feed-forward network with learnable edge-wise nonlinear functions from Kolmogorov-Arnold Networks; modelled stock-return patches through self-attention and integrated Fama-French three-factor and daily news-sentiment data through cross-attention.
- Built a PyTorch pipeline with 6,100 rolling samples from 30 U.S. stocks and four exogenous variables spanning 1998–2023; trained the model end to end under a negative Sharpe-ratio objective to generate long-only, fully invested portfolios, using Selective Learning to filter uncertain and anomalous observations.
- Achieved an out-of-sample Sharpe ratio of 1.375, above the JKP Common Task Framework leaderboard average of 1.321; ablations showed that removing KAN, Selective Learning, or both reduced the Sharpe ratio by 11.35%, 18.98%, and 25.02%, respectively. [Paper](#)

Economic Effects of LA 2028 Olympics

08/2025-03/2026

Position: Research Assistant, supervised by Dr. Lorena Martin, Trojans Sports Research Lab, USC

- Participating in a research project on analysing the effect of LA 2028 Olympics to local economy.

Corporate Carbon Finance Product Innovation and Pricing

03/2025-07/2025

Position: Research Assistant

Sponsor: International Institute of Green Finance (IIGF), CUFU, and State Grid Research Institute

- Contributed quantitative finance analysis to product-pricing discussions through weekly research meetings.

Platform Economy Research

09/2024-07/2025

Position: Research Assistant

Sponsor: Professor Shen Bo's Research Group, Wuhan University

- Collected market materials and analysed platform business models, competition, user behaviour, effects of advertising and its implications for consumer and social welfare.

Teaching Assistant for Game Theory Module in Wuhan University

02/2025-07/2025

Position: Teaching Assistant

- Assisted instructor in Game Theory course, collected and graded assignments monthly, selected as an outstanding TA, earning high recognition from both the instructor and students.
- Provided in-depth explanations during exercise classes, including addressing students' questions on Cournot equilibrium and other core models to strengthen their understanding of theoretical concepts.
- Organised Q & A sessions before quizzes and exams; supervised and graded mid-term and final exams, as well as collecting and grading monthly assignments.

Industry Experience (Internships)**China Telecom BestPay, Paid, Full-Time**

07/2026-09/2026

Position: Intern, Technology and Big Data Platform Department

- Engineered reusable LLM Skills that automatically ingest, structure, and update financial-violation records in a searchable database.
- Developed an enterprise risk model that combines LLM-derived violation signals with transaction, behavioural, and financial data to automate company risk classification.

Industrial and Commercial Bank of China, Paid, Full-Time

08/2025-10/2025

Position: Intern, Corporate Finance Department & International Business Department

- Assessed corporate credit applications and trade-finance documentation to identify market, financial, and transaction risks, supporting loan approval and cross-border financing decisions.

China Securities Co., Ltd, Full-Time

07/2024-08/2024

Position: Intern, Bond Underwriting Department

- Automated weekly bond-report data processing with Python and Wind, improving production efficiency by 40% and delivering 5 issues for market and partnership analysis.

KPMG Huazhen LLP, Paid, Full-Time

01/2024-02/2024

Position: Intern, Audit Department

- Built Excel templates for CRRC's annual audit to automate data import and validation, improving team efficiency by 70% while supporting evidence reconciliation and risk review.

Languages

Mandarin Chinese (native), English (proficient)

IELTS: Overall: 7.5 (Listening: 7.5, Reading: 9.0, Writing: 6.5, Speaking: 7.5)

Programming Skills

Python, The R Programming Language, STATA, LaTeX, MATLAB, Excel

Project Experience**Self-developed WeChat Mini Program**

A badminton-focused mini program built independently for match recording, Elo ratings, ranking comparison, and performance trend tracking. Currently have more than 700 users.

**Honours & Awards**

- Outstanding Student, Hongyi Honor College (2024–2025)

- Second-class Scholarship (2024–2025); Third-class Scholarship (2023–2024)

- Third Prize, National ETF Simulation Investment Elite Challenge

- Outstanding Graduation Thesis (2025–2026)

- Two-time Hongyi badminton team champion; runner-up in men's and mixed doubles

- University football: third place and MVP; college football: champion, MVP, and Golden Glove